

Feature form and antisymmetry augmentation for $\Delta\Delta G$ prediction

Experiment 07 · ddG_with_Boltz · raw pair-track (z) features · MLP, protein-holdout.

Two design choices for a $\Delta\Delta G$ regressor built on Boltz pair-track embeddings are settled by a within-dataset ablation (Tsuboyama and FireProt, protein-holdout). **Concatenating the wild-type and mutant pooled pair-vectors** (rather than only their difference) is consistently as good or better, and **antisymmetry augmentation** ($\Delta\Delta G(A\rightarrow B) = -\Delta\Delta G(B\rightarrow A)$) improves FireProt — but only in the concat representation, where the reverse mutation is a natural input transform. The two together are adopted as the default.

1. Motivation

A $\Delta\Delta G$ predictor on Boltz-2 pair-track (z) embeddings can be built from the mutated residue's pooled pair-vector in two ways: the **difference** of mutant and wild-type (Δz), or their **concatenation** (`concat = [wt, mut]`), which keeps both absolute levels; the difference is recoverable as their subtraction but not vice-versa). Separately, $\Delta\Delta G$ obeys an exact antisymmetry — the reverse mutation has the negated value — which can be injected as a training augmentation. We ask, on each dataset independently, whether either helps.

2. Methods

For a mutation at residue i , the pooled pair-vector is the mean over partners of the z row at i (128-dim), taken for the wild-type and mutant structures. Δz = the difference (with the diagonal element); **concat** = the two pooled vectors side by side. Both are 256-dim. **Antisymmetry augmentation** adds each reversed mutation (negated $\Delta\Delta G$) to the training set — for Δz by negating the feature vector, for concat by swapping the two halves. Evaluation is a protein-holdout (5-fold GroupKFold by protein), out-of-fold pooled metrics, with a 5-seed MLP ensemble. Corpora: Tsuboyama (12,359 mutations / 412 proteins) and FireProt (1,543 / 85).

3. Results

Dataset	Feature	no augmentation	+ antisymmetry
Tsuboyama	Δz (difference)	0.792	0.598
Tsuboyama	concat	0.801	0.799
FireProt	Δz (difference)	0.575	0.579
FireProt	concat	0.508	0.588

Table 1. Pooled Pearson r (protein-holdout). Best per dataset in bold.

07 — feature form × symmetry augmentation (FireProt = ≤ 500 , 3205/138)

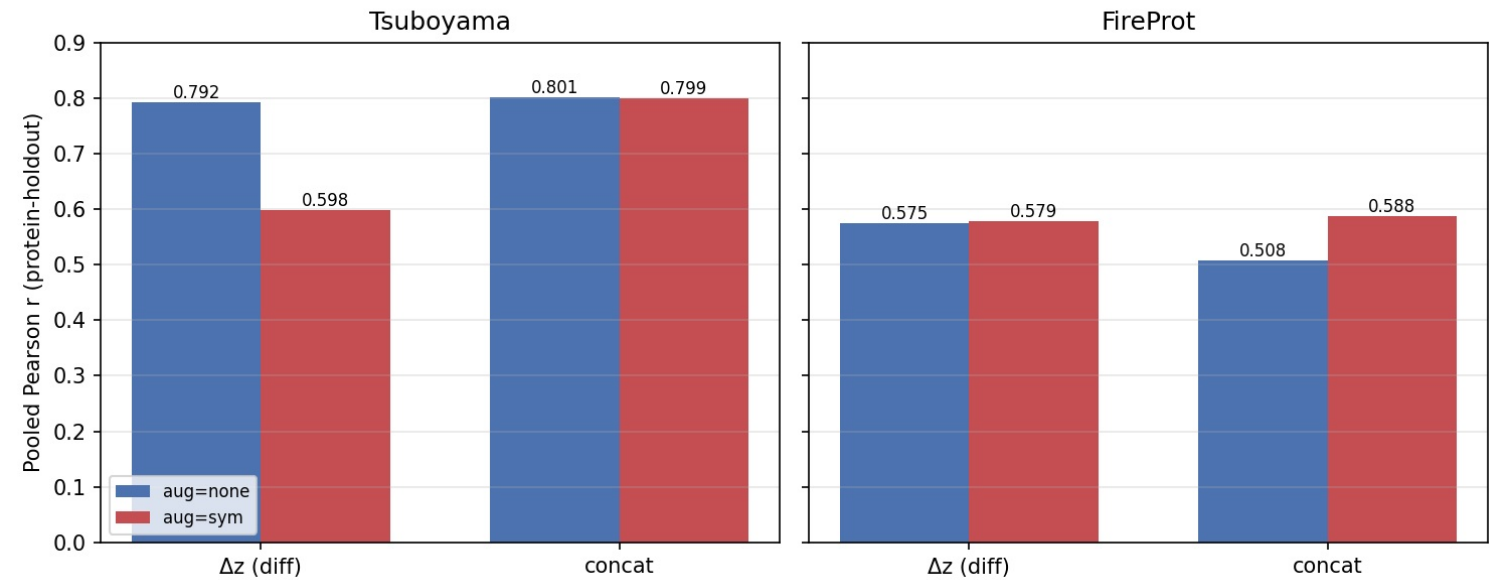


Figure 1. Protein-holdout Pearson r for the 2×2 ablation, per dataset.

4. Discussion

Concat $\geq \Delta z$ everywhere. Keeping both wild-type and mutant levels is consistently at least as good as the difference (Tsuboyama 0.792→0.801, FireProt 0.575→0.508) — a free gain, since concat strictly contains the difference.

Antisymmetry augmentation is representation-dependent. On Δz it forces an odd-function model ($f(-x) = -f(x)$) that fights a dataset skewed toward destabilizing mutations, collapsing Tsuboyama's calibration (Pearson 0.792→0.598) while rank correlation is preserved. In the concat representation the reverse mutation is instead a swap of the two halves — a valid input point — so augmentation is safe on Tsuboyama (0.801→0.799) and **improves FireProt** (0.508→0.588).

Conclusion. Use **concat features with antisymmetry augmentation**: neutral on Tsuboyama, a real gain on FireProt (+0.014 Pearson over the plain-difference baseline).

